Introduction To Industry 4.0 And Industrial Internet Of Things

Assignment-12

Number of questions: 15 **Total marks: 15 X 1= 15**

1. What are the advantages of implementing IoT in the chemical industry?

    a. Increase Energy Expenses
    b. Maximize Supply Chain Risk
    c. Improve Logistics
    d. Deteriorate Quality and equipment management

Answer: (c)

Justification: Please refer to Slide 11 of Lecture 1 of Week 12

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2. Which of the following is one of the primary goals of predictive maintenance?

    a. Maximizing equipment breakdowns
    b. Addressing historical issues only
    c. Reducing equipment breakdown
    d. Ignoring real-time problems

Answer: (c)

Justification: Please refer to Slide 12 of Lecture 1 of Week 12

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3. What is/are the benefits of using IoT sensors in the Pharmaceutical Industry?

    a. Access huge data of different manufacturing departments
    b. Real time monitoring
    c. Reduce production cost and wastage
    d. All of the above

Answer: (d)

Justification: Please refer to Slide 18 of Lecture 1 of Week 12

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4. “By applying IoT in the pharmaceutical industry, it is possible to detect Adverse Drugs Reaction (ADR), effects of pharmaceutical excipients, and allergies.” State true or false.

a. True
b. False

Answer: (a)

Justification: Please refer to Slide 19 of Lecture 1 of Week 12

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5. UAVs are capable of performing which of the following operations?

a. Aerial imagery
b. Visual imagery
c. Thermal imagery
d. All of the above

Answer: (d)

Justification: Please refer to Slide 3 of Lecture 2 of Week 12

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6. Which of the following is true about the first generation of the UAV technology?

a. Fundamental Remote-Control UAVs of different forms
b. Enhanced intelligent piloting models and full autonomy, full airspace awareness, auto action
c. Transformable designs with 360° gimbals, high quality video or higher-value instrumentation, improved piloting modes
d. Improved safety and regulatory, platform and payload adaptability, automated safety modes, intelligent piloting models and full autonomy, airspace awareness

Answer: (a)

Justification: Please refer to Slide 4 of Lecture 2 of Week 12

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7. How is the application of UAVs in the construction site beneficial?

a. It does not show clients work progress when they are far away from job sites
b. Require more time, energy and money
c. Monitor shoreline erosion
d. All of the above

Answer: (c)

Justification: Please refer to Slide 9 of Lecture 2 of Week 12

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8. What are the advantages of using UAVs in telecommunication?

a. Monitor towers from any angle and height
b. Broadcasting only historical events
c. Decrease work safety
d. Covers local areas with high cost

Answer: (a)

Justification: Please refer to Slide 13,14 of Lecture 2 of Week 12

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9. “By using UAVs it is possible to scan a huge number of items in a warehouse, check the missing items and monitoring full inventory in a day.” State true or false.

a. True
b. False

Answer: (a)

Justification: Please refer to Slide 17 of Lecture 2 of Week 12

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10. Which of the following can be performed by using UAVs in the forestry?

a. Manage forest plantations and evenly distribute seedlings sprinkling fertilizer
b. Measure the carbon storage in biomass by remote sensing
c. Resist deforestation and increase security
d. All of the above

Answer: (d)

Justification: Please refer to Slide 19 of Lecture 2 of Week 12

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11. Which of the following activities can be performed by using UAVs in the mining?

a. Regular surface survey for optimized blast design
b. Manage stockpiles
c. Site exploration
d. All of the above

Answer: (d)

Justification: Please refer to Slide 11 of Lecture 2 of Week 12

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12. “Case study enables a researcher to closely examine the data within a specific context by following certain procedures and providing quantitative and qualitative analysis of the data.” State true or false.

a. True
b. False

Answer: (a)

Justification: Please refer to Slide 2 of Lecture 3 of Week 12

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13. What is the primary purpose of case studies?

a. Create fictional scenarios for analysis
b. Explore and investigate real-life phenomena
c. Focus on large geographical areas
d. Involve a vast number of individuals in the study

Answer: (b)

Justification: Please refer to Slide 3 of Lecture 3 of Week 12

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14. What does IoD stand for in the context of IoT and IIoT?

a. Internet of Devics
b. Internet of Drones
c. Intensity of Data
d. Intention of Damage

Answer: (b)

Justification: Please refer to Page 308 of the book Introduction to IoT by S Misra, A Mukherjee, A Roy by Cambridge University Press

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15. Which of the following is false about the application of UAVs in agriculture?

a. Increase effective yields
b. Crop health monitoring
c. Hinder farmers in scouting their crops
d. Optimize use of seed, fertilizer, water

Answer: (c)

Justification: Please refer to Slide 6 of Lecture 2 of Week 12
